

Date: _____

Assignments-01

Name :- Ahmed Naseer
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Subject :- Data Structures
Program :- BS DS 3-1

v Project Description:-

The FinTech Trading Management System is a Java application built for managing an online investment platform. It handles market asset tracking, linear and binary searches, multiple array sorting algorithms, custom watchlist maintenance using linked list, recursive portfolio valuation and transaction/order handling using stacks and queues.

ii Data Structures and Algorithms used:-

- Arrays:- Stores the array of 10 market assets (Asset IDs: 201 to 210) containing name, asset type, price and percentage change.
- Linear Search: Scans element sequentially to locate assets by ID.
- Binary Search: Find assets by ID in $O(\log n)$ time after sorting.
- Bubble Sort: Sorts assets by current price in ascending order.
- Selection Sort:- Sorts assets based on percentage change.
- Insertion Sort:- Orders assets sequentially by Asset ID.
- Merge Sort: Divide-and-conquer algorithm to sort assets by price.

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- Quick Sort: Sorts Assets by Asset ID using Lomuto Partitioning.
- Linked List: A singly linked list managing dynamic insertion, deletion, and searching of watchlist items.

Recursion: Calculates total portfolio valuation and displays holding subtotals without loops.

Stack: Tracks recent buy/sell transactions following Last-In, First-Out (LIFO).

Queue: Processes market trading orders in First-In, First-Out (FIFO) sequence.

3) Dry Runs:

Dry Run 1: Binary Search (Searching for ID = 205)

- Sorted Array IDs: [201, 202, 203, 204, 205, 206, 207, 208, 209, 210]

- Pointers: low = 0, high = 9, target = 205

- Iteration 1:

- $mid = 0 + \frac{9-0}{2} = 4$

- $Array[4].id = 205$

- Target match found immediately at index 4!

Dry Run 2: Recursive Portfolio Calculation (first 2 items)

Assets: Cardano (Price = 0.45, Qty = 1000), Google (Price = 175.50, Qty = 20)

- Recursive Sequence:-

- $calculate(assets, qty, 0) \rightarrow 0.45 \times 1000 + calculate(assets, qty, 1)$

- $calculate(assets, qty, 1) \rightarrow (175.50 \times 20) + calculate(assets, qty, 2)$

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- Subtotals:
 - Cardano : \$450.00
 - Google : \$3,510.00
 - Cumulative Subtotal Indices 0 to 11: \$3,960.00

Dry Run 3: Stack operations (LIFO):

1. push ("Buy 50. GOOGLE")
Stack: ["SELL 100 AMD", "BUY 50 GOOGLE"]
2. push ("SELL 100 AMD")
Stack: ["SELL 100 AMD", "BUY 50 GOOGLE"]
- 3) pop()
 - Removed : "SELL 100 AMD"
 - Stack : ["BUY 50 GOOGLE"]

4) Sample Output:..

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FINTECH TRADING MANAGEMENT SYSTEM

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1. Manage Market Assets
2. Search Asset
3. Set Market Asset.
4. Manage Watchlist.
5. Calculate Portfolio Value.
6. Manage Transactions.
7. Managing Trading Orders.
8. Exit

Enter your choice: 2

--- SEARCH ASSET ---

Date: _____

1. Linear Search (by Asset ID)
2. Binary Search (by Asset ID)
0. Back to Main Menu

Enter your choice: 2

Enter Asset ID to search: 205

Sorting assets by ID for Binary Search...

[Tracing] Low: 0 | Mid: 4 | High: 9 | Checking ID: 205

Asset found at index 4 using Binary Search:

ID: 205 | Name: AMD | Type: Stock | Price: \$160.00 // Change

--- Portfolio Details ---

--- PORTFOLIO DETAILS ---

Holding: Cardano | Price: \$0.45 | Qty: 1000 | Subtotal: \$450.00

Holding: Google | Price: \$175.50 | Qty: 20 | Subtotal: \$3510.00

Holding: Silver | Price: \$28.40 | Qty: 100 | Subtotal: \$2840.00

Holding: Polkadot | Price: \$6.20 | Qty: 500 | Subtotal: \$3100.00

Total Calculated Portfolio Value (Recursive): \$23854.50

Sample Output:

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FintechTradingSystem (run) ×  FintechTradingSystem (run) #2 ×

run:

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      FINTECH TRADING MANAGEMENT SYSTEM
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1. Manage Market Assets
2. Search Asset
3. Sort Market Assets
4. Manage Watchlist
5. Calculate Portfolio Value
6. Manage Transactions
7. Manage Trading Orders
0. Exit
Enter your choice: 1

--- MARKET ASSETS ---
1. Display All Assets
2. Calculate Average Price
3. Find Highest-Priced Asset
4. Find Lowest-Priced Asset
0. Back to Main Menu
Enter your choice: 1

--- All Market Assets ---
ID: 201 | Name: Cardano      | Type: Crypto   | Price: $0.45   | Change: +3.20%
ID: 202 | Name: Google             | Type: Stock    | Price: $175.50 | Change: -1.10%
ID: 203 | Name: Silver              | Type: Commodity| Price: $28.40  | Change: +1.80%
ID: 204 | Name: Polkadot            | Type: Crypto   | Price: $6.20   | Change: -0.50%
ID: 205 | Name: AMD                 | Type: Stock    | Price: $160.00 | Change: +5.40%
ID: 206 | Name: S&P 500             | Type: Index    | Price: $5450.00| Change: +0.70%
ID: 207 | Name: Intel               | Type: Stock    | Price: $31.20  | Change: -2.80%
ID: 208 | Name: Ripple              | Type: Crypto   | Price: $0.58   | Change: +4.00%
ID: 209 | Name: Netflix             | Type: Stock    | Price: $680.30 | Change: +2.10%
ID: 210 | Name: Copper              | Type: Commodity| Price: $4.15   | Change: -0.90%

--- MARKET ASSETS ---
1. Display All Assets
2. Calculate Average Price
3. Find Highest-Priced Asset
4. Find Lowest-Priced Asset
0. Back to Main Menu
Enter your choice: |
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